

Phys 115B HW 2

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Question 1. *More Practice with Spherical Harmonics*

Consider a particle in three dimensions in a state

$$\psi(x, y, z) = A(x + y + 2z)e^{-\alpha r}$$

where $\alpha > 0$ is a positive constant, and A a normalization factor.

(a) *Write the spherical harmonics $Y_1^{0,\pm 1}$ in terms of x, y, z , and*

$$r = \sqrt{x^2 + y^2 + z^2}.$$

(b) *Using your result from part (a), show that ψ can be written as a product of a purely r -dependent wave function and a linear combination of spherical harmonics.*

Proof. First, recall that the spherical harmonics are given as follow:

$$Y_\ell^m = \epsilon \sqrt{\frac{2\ell+1}{4\pi} \frac{(\ell-|m|)!}{(\ell+|m|)!}} e^{im\phi} P_\ell^m(\cos(\theta)), \quad \epsilon = \begin{cases} (-1)^m & m \leq 0 \\ 1 & m > 0 \end{cases} \quad (1)$$

$$P_\ell^m(z) = (1-z^2)^{|m|/2} \left(\frac{d}{dz}\right)^{|m|} \left(\frac{1}{2^\ell \ell!} \left(\frac{d}{dz}\right)^\ell (z^2-1)^\ell\right) \quad (2)$$

Where P_ℓ^m is the associated Legendre's Polynomial.

(a) Given $\ell = 1$ and $m = 0, \pm 1$, the associated spherical harmonics (in spherical coordinates) are as follow:

$$\ell = 1, m = 0 \implies P_1^0(z) = \frac{1}{2} \frac{d}{dz} (z^2 - 1) = z \quad (3)$$

$$\implies Y_1^0 = \sqrt{\frac{3}{4\pi}} \cos(\theta) \quad (4)$$

$$\ell = 1, m = 1 \implies P_1^1(z) = \sqrt{1-z^2} \frac{d}{dz} \left(\frac{1}{2} \frac{d}{dz} (z^2 - 1)\right) = \sqrt{1-z^2} \quad (5)$$

$$\implies Y_1^1 = \sqrt{\frac{3}{8\pi}} e^{i\phi} \sqrt{1-\cos^2(\theta)} = \sqrt{\frac{3}{8\pi}} e^{i\phi} \sin(\theta) \quad (6)$$

$$\ell = 1, m = -1 \implies P_1^{-1}(z) = \sqrt{1-z^2} \frac{d}{dz} \left(\frac{1}{2} \frac{d}{dz} (z^2 - 1) \right) = \sqrt{1-z^2} \quad (7)$$

$$\implies Y_1^{-1} = -\sqrt{\frac{3}{8\pi}} e^{-i\phi} \sqrt{1 - \cos^2(\theta)} = -\sqrt{\frac{3}{8\pi}} e^{-i\phi} \sin(\theta) \quad (8)$$

Then, using the cartesian coordinates $x = r \sin(\theta) \cos(\phi)$, $y = r \sin(\theta) \sin(\phi)$, $z = r \cos(\theta)$, and $r = \sqrt{x^2 + y^2 + z^2}$, the above becomes the following:

$$Y_1^0 = \sqrt{\frac{3}{4\pi}} \cos(\theta) = \sqrt{\frac{3}{4\pi}} \frac{z}{r} \quad (9)$$

$$Y_1^1 = \sqrt{\frac{3}{8\pi}} (\cos(\phi) + i \sin(\phi)) \sin(\theta) = \sqrt{\frac{3}{8\pi}} \frac{x + iy}{r} \quad (10)$$

$$Y_1^{-1} = -\sqrt{\frac{3}{8\pi}} (\cos(\phi) - i \sin(\phi)) \sin(\theta) = \sqrt{\frac{3}{8\pi}} \frac{-x + iy}{r} \quad (11)$$

(b) Using the equations (9),(10),(11) from part (a), the linear combinations of $Y_1^{\pm 1}$ provides the following:

$$Y_1^1 + Y_1^{-1} = \sqrt{\frac{3}{8\pi}} \frac{2iy}{r} \implies y = -ir \sqrt{\frac{2\pi}{3}} (Y_1^1 + Y_1^{-1}) \quad (12)$$

$$Y_1^1 - Y_1^{-1} = \sqrt{\frac{3}{8\pi}} \frac{2x}{r} \implies x = r \sqrt{\frac{2\pi}{3}} (Y_1^1 - Y_1^{-1}) \quad (13)$$

$$Y_1^0 = \sqrt{\frac{3}{4\pi}} \frac{z}{r} \implies z = \sqrt{2} r \sqrt{\frac{2\pi}{3}} Y_1^0 \quad (14)$$

Hence, the term $x + y + 2z$ can be rewritten as:

$$x + y + 2z = r \sqrt{\frac{2\pi}{3}} (Y_1^1 - Y_1^{-1}) - ir \sqrt{\frac{2\pi}{3}} (Y_1^1 + Y_1^{-1}) + \sqrt{2} r \sqrt{\frac{2\pi}{3}} Y_1^0 \quad (15)$$

$$= r \sqrt{\frac{2\pi}{3}} ((Y_1^1 - Y_1^{-1}) - i(Y_1^1 + Y_1^{-1}) + \sqrt{2} Y_1^0) \quad (16)$$

$$= r \sqrt{\frac{2\pi}{3}} ((1-i)Y_1^1 - (1+i)Y_1^{-1} + \sqrt{2} Y_1^0) \quad (17)$$

So, the proposed ψ can be given as follow:

$$\psi = A(x + y + 2z)e^{-\alpha r} \quad (18)$$

$$= A \sqrt{\frac{2\pi}{3}} ((1-i)Y_1^1 - (1+i)Y_1^{-1} + \sqrt{2} Y_1^0) \cdot r e^{-\alpha r} \quad (19)$$

Which, this is a product of purely radial-dependent function ($r e^{-\alpha r}$), together with linear combinations of the spherical harmonics $((1-i)Y_1^1 - (1+i)Y_1^{-1} + \sqrt{2} Y_1^0)$, up to some scalars.

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Question 2. Radial Momentum

In lecture, we defined an operator corresponding to the momentum in the radial direction,

$$p_r \equiv \frac{1}{2}(\hat{\mathbf{r}} \cdot \mathbf{p} + \mathbf{p} \cdot \hat{\mathbf{r}}) = -\frac{i\hbar}{2} \left(\frac{1}{r} \mathbf{r} \cdot \nabla + \nabla \cdot \frac{\mathbf{r}}{r} \right).$$

(a) Show that $\hat{\mathbf{r}} \cdot \mathbf{p}$ is not Hermitian, and hence cannot itself be identified with momentum in the radial direction.

(b) Using the definition of p_r above, show that

$$p_r^2 = -\hbar^2 \frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right).$$

Proof. First, recall the definition of these operators:

$$\hat{\mathbf{r}} = \frac{x}{r}\hat{x} + \frac{y}{r}\hat{y} + \frac{z}{r}\hat{z}, \quad \mathbf{p} = p_x\hat{x} + p_y\hat{y} + p_z\hat{z} \quad (20)$$

(Note: Here for $u = x, y, z$ the expression $\hat{x}$ is representing coordinates, rather than operators; the x, y, z and the partials are the operators).

(a) Given $\hat{\mathbf{r}} \cdot \mathbf{p}$, it is given as follow:

$$\hat{\mathbf{r}} \cdot \mathbf{p} = \left(\frac{x}{r}\hat{x} + \frac{y}{r}\hat{y} + \frac{z}{r}\hat{z} \right) \cdot (p_x\hat{x} + p_y\hat{y} + p_z\hat{z}) = \frac{1}{r}(xp_x + yp_y + zp_z) \quad (21)$$

Which, the Hermitian conjugate (adjoint) of this is given as follow:

$$(\hat{\mathbf{r}} \cdot \mathbf{p})^\dagger = \frac{1}{r}(xp_x + yp_y + zp_z)^\dagger = \frac{1}{r}(p_x x + p_y y + p_z z) \quad (22)$$

And, given the commutation relation $[u, p_u] = -i\hbar$, one has $p_u u = u p_u + i\hbar$. This simplifies to the following:

$$(\hat{\mathbf{r}} \cdot \mathbf{p})^\dagger = \frac{1}{r}(p_x x + p_y y + p_z z) = \frac{1}{r}(xp_x + yp_y + zp_z) + \frac{3i\hbar}{r} = \hat{\mathbf{r}} \cdot \mathbf{p} + \frac{3i\hbar}{r} \quad (23)$$

This shows that $\hat{\mathbf{r}} \cdot \mathbf{p}$ is not Hermitian (self-adjoint).

(b) First, do the calculation of $\mathbf{p} \cdot \hat{\mathbf{r}}$, one yields the following:

$$\mathbf{p} \cdot \hat{\mathbf{r}} = (p_x\hat{x} + p_y\hat{y} + p_z\hat{z}) \cdot \left(\frac{x}{r}\hat{x} + \frac{y}{r}\hat{y} + \frac{z}{r}\hat{z} \right) \quad (24)$$

$$= -i\hbar \left(\frac{\partial}{\partial x} \left(\frac{x}{r} \right) + \frac{\partial}{\partial y} \left(\frac{y}{r} \right) + \frac{\partial}{\partial z} \left(\frac{z}{r} \right) \right) \quad (25)$$

$$= -i\hbar \left(\frac{3}{r} - \frac{x^2 + y^2 + z^2}{r^3} + \frac{1}{r} \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} + z \frac{\partial}{\partial z} \right) \right) \quad (26)$$

$$= -\frac{2i\hbar}{r} - \frac{i\hbar}{r} \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} + z \frac{\partial}{\partial z} \right) \quad (27)$$

$$= -\frac{2i\hbar}{r} + \frac{1}{r}(xp_x + yp_y + zp_z) \quad (28)$$

Also, notice that given the definition $r := \sqrt{x^2 + y^2 + z^2}$, one has the following regarding the partial derivatives:

$$u = x, y, z, \quad \frac{\partial}{\partial u} = \frac{\partial r}{\partial u} \frac{\partial}{\partial r} = \frac{u}{\sqrt{x^2 + y^2 + z^2}} \frac{\partial}{\partial r} = \frac{u}{r} \frac{\partial}{\partial r} \quad (29)$$

Hence, regarding the full definition, one has the following:

$$p_r = \frac{1}{2}(\hat{\mathbf{r}} \cdot \mathbf{p} + \mathbf{p} \cdot \hat{\mathbf{r}}) \quad (30)$$

$$= \frac{1}{2} \left(\frac{1}{r}(xp_x + yp_y + zp_z) - \frac{2i\hbar}{r} + \frac{1}{r}(xp_x + yp_y + zp_z) \right) \quad (31)$$

$$= \frac{1}{2} \left(-\frac{2i\hbar}{r} \left(x \frac{\partial}{\partial x} + y \frac{\partial}{\partial y} + z \frac{\partial}{\partial z} \right) - \frac{2i\hbar}{r} \right) \quad (32)$$

$$= \frac{1}{2} \left(-\frac{2i\hbar}{r} \cdot \frac{x^2 + y^2 + z^2}{r} \frac{\partial}{\partial r} - \frac{2i\hbar}{r} \right) \quad (33)$$

$$= -i\hbar \left(\frac{\partial}{\partial r} + \frac{1}{r} \right) \quad (34)$$

Hence, the operator p_r^2 provides the following:

$$p_r^2 = -\hbar^2 \left(\frac{\partial}{\partial r} + \frac{1}{r} \right)^2 \quad (35)$$

$$= -\hbar^2 \left(\frac{\partial^2}{\partial r^2} - \frac{1}{r^2} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r} \frac{\partial}{\partial r} + \frac{1}{r^2} \right) \quad (36)$$

$$= -\hbar^2 \left(\frac{\partial^2}{\partial r^2} + \frac{2}{r} \frac{\partial}{\partial r} \right) \quad (37)$$

$$= -\frac{\hbar^2}{r^2} \left(r^2 \frac{\partial^2}{\partial r^2} + 2r \frac{\partial}{\partial r} \right) \quad (38)$$

$$= -\frac{\hbar^2}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial}{\partial r} \right) \quad (39)$$

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Question 3. *The Size of Hydrogen (Griffiths 4.15, modified, 4.13 in second edition, still modified). Do parts (a)–(c) from the Griffiths problem.*

(a) Find $\langle r \rangle$ and $\langle r^2 \rangle$ for an electron in the ground state of hydrogen. Express your answers in terms of the Bohr radius.

(b) Find $\langle x \rangle$ and $\langle x^2 \rangle$ for an electron in the ground state of hydrogen.

(Hint: This requires no new integration - note that $r^2 = x^2 + y^2 + z^2$, and exploit the symmetry of the ground state).

(c) Find $\langle x^2 \rangle$ in the state $n = 2, \ell = 1, m = 1$.

(Hint: this state is not symmetrical in x, y, z . Use $x = r \sin(\theta) \cos(\phi)$).

(d) Find $\langle r \rangle$ and $\langle r^2 \rangle$ for an electron in a circular orbit of hydrogen with arbitrary principal quantum number n (corresponds to $\ell = n - 1$ and any allowed m).

(e) Compute the RMS uncertainty

$$\sqrt{\langle r^2 \rangle - \langle r \rangle^2}$$

in terms of r for the electron in part (d). Note that the fractional spread in r decreases with increasing n (in this sense the system “begins to look classical” for large n). How much more volume does a hydrogen atom in the $n = 100$ state occupy compared to the hydrogen atom in the ground state. (Hint — you might want to look at Griffiths 4.55, or 4.15 in the second edition.)

Proof. Recall that the ground state of hydrogen has the following wavefunction:

$$\psi_{100} = \frac{1}{\sqrt{\pi a^3}} e^{-r/a} \quad (40)$$

Where a represents the Bohr radius.

(a) To calculate the expectation values, one has the following:

$$\langle r \rangle = \int_{\mathbb{R}^3} r |\psi|^2 dV = \int_{r=0}^{\infty} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{r}{\pi a^3} e^{-2r/a} r^2 \sin(\theta) d\phi d\theta dr \quad (41)$$

$$= \frac{2\pi}{\pi a^3} \left(\int_0^{\infty} r^3 e^{-2r/a} dr \right) \left(\int_0^{\pi} \sin(\theta) d\theta \right) \quad (42)$$

$$= -\frac{4}{a^3} \left(\frac{a}{2} r^3 + \frac{3a^2}{4} r^2 + \frac{3a^3}{4} r + \frac{3a^4}{8} \right) e^{-2r/a} \Big|_0^{\infty} \quad (43)$$

$$= \frac{4}{a^3} \cdot \frac{3a^4}{8} = \frac{3a}{2} \quad (44)$$

$$\langle r^2 \rangle = \int_{\mathbb{R}^3} r^2 |\psi|^2 dV = \int_{r=0}^{\infty} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{r^2}{\pi a^3} e^{-2r/a} r^2 \sin(\theta) d\phi d\theta dr \quad (45)$$

$$= \frac{2\pi}{\pi a^3} \left(\int_0^{\infty} r^4 e^{-2r/a} dr \right) \left(\int_0^{\pi} \sin(\theta) d\theta \right) \quad (46)$$

$$= -\frac{4}{a^3} \left(\frac{a}{2} r^4 + a^2 r^3 + \frac{3a^3}{2} r^2 + \frac{3a^4}{2} r + \frac{3a^5}{4} \right) e^{-2r/a} \Big|_0^{\infty} \quad (47)$$

$$= \frac{4}{a^3} \cdot \frac{3a^5}{4} = 3a^2 \quad (48)$$

- (b) First, for $\langle x^2 \rangle$, since ψ is under spherical symmetry, this implies that the expectation value of x^2, y^2, z^2 will all be the same; furthermore, since $r^2 = x^2 + y^2 + z^2$, one has $\langle r^2 \rangle = \langle x^2 \rangle + \langle y^2 \rangle + \langle z^2 \rangle = 3 \langle x^2 \rangle$ based on the symmetry claim. Hence, with $\langle r^2 \rangle = 3a^2$, one has $\langle x^2 \rangle = a^2$.

Now, to calculate $\langle x \rangle$, notice that based on the spherical symmetry, one has $\psi(x, y, z) = \psi(-x, y, z)$. Hence, when fixing y, z , one has ψ being an even function with respect to x (and so is $|\psi|^2$). Doing the expectation value, one has the following:

$$\langle x \rangle = \int_{\mathbb{R}^3} x |\psi|^2 dV = \int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} \left(\int_{-\infty}^{\infty} x |\psi|^2 dx \right) dy \right) dz = 0 \quad (49)$$

And, the reason is because $\int_{-\infty}^{\infty} x |\psi|^2 dx = 0$ due to the parity of the function.

- (c) Given $n = 2, \ell = 1, m = 1$ (where $\ell = n - 1$), one has the wavefunction $\psi = R_{n,n-1} Y_1^1$.

Using the lecture notes (together with the calculation in **Problem 1** in this HW), we get $R_{2,1} = \frac{1}{\sqrt{4!}} \frac{1}{a_0^{3/2}} \frac{r}{a_0} e^{-r/(2a_0)}$, and $Y_1^1 = \sqrt{\frac{3}{8\pi}} e^{i\phi} \sin(\theta)$. Then, the expectation value of x becomes the following:

$$\langle x^2 \rangle = \int_{\mathbb{R}^3} x^2 |\psi|^2 dV = \int_{r=0}^{\infty} \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} r^2 \sin^2(\theta) \cos^2(\phi) \cdot \frac{r^2}{24a_0^5} e^{-r/(a_0)} \cdot \frac{3}{8\pi} \sin^2(\theta) \cdot r^2 \sin(\theta) d\phi d\theta dr \quad (50)$$

$$= \frac{1}{64\pi a_0^5} \left(\int_{r=0}^{\infty} r^6 e^{-r/(a_0)} dr \right) \left(\int_0^{2\pi} \cos^2(\phi) d\phi \right) \left(\int_0^{\pi} \sin^5(\theta) d\theta \right) \quad (51)$$

Which, doing individual calculation, one yields:

$$\int_0^{2\pi} \cos^2(\phi) d\phi = \int_0^{2\pi} \frac{1 + \cos(2\phi)}{2} d\phi = \pi \quad (52)$$

$$\int_0^{\pi} \sin^5(\theta) d\theta = - \int_0^{\pi} (1 - 2\cos^2(\theta) + \cos^4(\theta)) (-\sin(\theta)) d\theta \quad (53)$$

$$= \int_{-1}^1 (1 - 2u^2 + u^4) du = \left(u - \frac{2}{3}u^3 + \frac{1}{5}u^5 \right) \Big|_{-1}^1 \quad (54)$$

$$= \frac{16}{15} \quad (55)$$

$$\int_0^{\infty} r^6 e^{-r/(a_0)} dr = \int_0^{\infty} a_0^7 u^6 e^{-u} du = a_0^7 \Gamma(7) = a_0^7 \cdot 6! \quad (56)$$

Combine the values, one gets:

$$\langle x^2 \rangle = \frac{1}{64\pi a_0^5} \cdot a_0^7 \cdot 6! \cdot \pi \cdot \frac{16}{15} = 12a_0^2 \quad (57)$$

- (d) Given arbitrary n , $\ell = n - 1$, and any allowed m (which $-\ell \leq m \leq \ell$). Since the spherical harmonics are essentially normalized (together with the expectation values only dependent on r , which the spherical harmonics are independent of this), one only needs to consider the case for the radial equation, which is given by $R_{n,n-1} = \frac{1}{\sqrt{(2n)!}} \left(\frac{2}{na_0}\right)^{3/2} \left(\frac{2r}{na_0}\right)^{n-1} e^{-r/(na_0)}$. Which, including the spherical coordinates' scaling factor, one gets the following:

$$\langle r \rangle = \int_0^\infty r |R_{n,n-1}|^2 r^2 dr = \frac{1}{(2n)!} \left(\frac{2}{na_0}\right)^{2n+1} \int_0^\infty r^{2n+1} e^{-2r/(na_0)} dr \quad (58)$$

$$= \frac{1}{(2n)!} \frac{na_0}{2} \int_0^\infty u^{2n+1} e^{-u} du = \frac{1}{(2n)!} \frac{na_0}{2} \Gamma(2n+2) = (2n+1) \frac{na_0}{2} \quad (59)$$

$$\langle r^2 \rangle = \int_0^\infty r^2 |R_{n,n-1}|^2 r^2 dr = \frac{1}{(2n)!} \frac{na_0}{2} \left(\frac{2}{na_0}\right)^{2n+2} \int_0^\infty r^{2n+2} e^{-2r/(na_0)} dr \quad (60)$$

$$= \frac{1}{(2n)!} \left(\frac{na_0}{2}\right)^2 \int_0^\infty u^{2n+2} e^{-u} du = \frac{\Gamma(2n+3)}{(2n)!} \left(\frac{na_0}{2}\right)^2 = (2n+1)(2n+2) \left(\frac{na_0}{2}\right)^2 \quad (61)$$

- (e) Given the answer in part (d), the uncertainty of r is given as follow (for each n , $\ell = n - 1$, and $-\ell \leq m \leq \ell$):

$$\sigma_r = \sqrt{\langle r^2 \rangle - \langle r \rangle^2} = \sqrt{(2n+1)(2n+2) \left(\frac{na_0}{2}\right)^2 - (2n+1)^2 \left(\frac{na_0}{2}\right)^2} \quad (62)$$

$$= \sqrt{2n+1} \frac{na_0}{2} = \frac{\langle r \rangle}{\sqrt{2n+1}} \quad (63)$$

Based on interpretation, the volume occupied could be interpreted as having the radius within $\pm\sigma_r$ of the expectation value $\langle r \rangle$.

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Question 4. *The Size of Hydrogen continued (Griffiths 4.16, 4.14 in the second edition)*

What is the most probable value of r , in the ground state of hydrogen? (The answer is not zero!)

Hint: First you must figure out the probability that the electron would be found between r and $r + dr$.

Proof. First, the ground state wavefunction of hydrogen atom is given as follow:

$$\psi_{100} = \frac{1}{\sqrt{\pi a^3}} e^{-r/a} \quad (64)$$

First, given any $R \geq 0$, the probability that the electron is at most radius R , one has the following:

$$\mathbb{P}(\leq R) = \int_{r=0}^R \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{1}{\pi a^3} e^{-2r/a} r^2 \sin(\theta) d\phi d\theta dr \quad (65)$$

$$= \frac{2\pi}{\pi a^3} \left(\int_{r=0}^R r^2 e^{-2r/a} dr \right) \left(\int_0^{\pi} \sin(\theta) d\theta \right) \quad (66)$$

$$= \frac{4}{a^3} \left(\int_{r=0}^R r^2 e^{-2r/a} dr \right) \quad (67)$$

Since this represents a cumulative probability function for $R \geq 0$, then the corresponding probability density with respect to radius R is:

$$p(R) = \frac{d\mathbb{P}(\leq R)}{dR} = \frac{4}{a^3} \frac{d}{dR} \left(\int_{r=0}^R r^2 e^{-2r/a} dr \right) = \frac{4}{a^3} R^2 e^{-2R/a} \quad (68)$$

Then, the most probable value of r is precisely the global maximum of $p(R)$ on $[0, \infty)$. Take the derivative's evaluation to be 0, one has the following:

$$\frac{dp(R)}{dR} = \frac{4}{a^3} \left(2R e^{-2R/a} - \frac{2}{a} R^2 e^{-2R/a} \right) = \frac{8}{a^4} R (a - R) e^{-2R/a} = 0 \quad (69)$$

Which, this is true iff $R = 0$ or $R = a$.

However, since $p(0) = 0$, the only possible maximum must be at $R = a$. Hence, it's most probable to have radius $r = a$, the Bohr radius. $\square$

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Question 5. *Where is the Electron (Griffiths 4.54, 4.45 in the second edition)*

What is the probability that an electron in the ground state of hydrogen will be found inside the nucleus?

- (a) First calculate the exact answer, assuming the wave function is correct all the way down to $r = 0$. Let b be the radius of the nucleus.

Here is the wave function:

$$\psi_{100}(r, \theta, \phi) = \frac{1}{\sqrt{\pi a^3}} e^{-r/a}$$

- (b) Expand your result as a power series in the small number $\epsilon := \frac{2b}{a}$, and show that the lowest-order term is the cubic: $p \approx \frac{4}{3} \left(\frac{b}{a}\right)^3$. This should be a suitable approximation, provided that $b \ll a$ (which it is).
- (c) Alternatively, we might assume that $\psi(r)$ is essentially constant over the (tiny) volume of the nucleus, so that $P \approx \frac{4}{3}\pi b^3 |\psi(0)|^2$. Check that you get the same answer this way.
- (d) Use $b \approx 10^{-15}$ m and $a \approx 0.5 \cdot 10^{-10}$ m to get a numerical estimate for P . Roughly speaking, this represents the “fraction of its time that the electron spends inside the nucleus”.

Proof.

- (a) To calculate the probability P that an electron will be found inside the nucleus (given it's in ground state) is given as follow (assuming the nucleus is of a solid ball with radius b):

$$P = \int_{\|\vec{r}\| \leq b} |\psi_{100}|^2 dV = \int_{r=0}^b \int_{\theta=0}^{\pi} \int_{\phi=0}^{2\pi} \frac{1}{\pi a^3} e^{-2r/a} r^2 \sin(\theta) d\phi d\theta dr \quad (70)$$

$$= \frac{2\pi}{\pi a^3} \left(\int_0^b r^2 e^{-2r/a} dr \right) \left(\int_0^{\pi} \sin(\theta) d\theta \right) = \frac{4}{a^3} \left(-\frac{a}{2} r^2 - \frac{a^2}{2} r - \frac{a^3}{4} \right) e^{-2r/a} \Big|_0^b \quad (71)$$

$$= 1 - \left(\frac{2b^2}{a^2} + \frac{2b}{a} + 1 \right) e^{-2b/a} \quad (72)$$

- (b) Define $\epsilon := \frac{2b}{a}$, then under the case $b \ll a$, one has $\epsilon \approx 0$. So, using the power series expansion, $e^{-2b/a} = e^{-\epsilon}$ has the form $e^{-\epsilon} = \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \epsilon^n$.

Plugin to the value derived in part (a), one yields the following:

$$P = 1 - \left(\frac{1}{2} \left(\frac{2b}{a} \right)^2 + \frac{2b}{a} + 1 \right) e^{-2b/a} = 1 - \left(\frac{\epsilon^2}{2} + \epsilon + 1 \right) e^{-\epsilon} \quad (73)$$

$$= 1 - \left(\frac{\epsilon^2}{2} + \epsilon + 1 \right) \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \epsilon^n \quad (74)$$

$$= 1 - \left(\sum_{n=0}^{\infty} \frac{(-1)^n}{2(n!)} \epsilon^{n+2} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \epsilon^{n+1} + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \epsilon^n \right) \quad (75)$$

$$= 1 - \left(\sum_{n=2}^{\infty} \frac{(-1)^{n-2}}{2((n-2)!)} \epsilon^n + \sum_{n=1}^{\infty} \frac{(-1)^{n-1}}{(n-1)!} \epsilon^n + \sum_{n=0}^{\infty} \frac{(-1)^n}{n!} \epsilon^n \right) \quad (76)$$

$$= 1 - \left(\sum_{n=2}^{\infty} (-1)^n \left(\frac{1}{2 \cdot (n-2)!} - \frac{1}{(n-1)!} + \frac{1}{n!} \right) \epsilon^n + \epsilon - \epsilon + 1 \right) \quad (77)$$

$$= \sum_{n=2}^{\infty} (-1)^{n+1} \left(\frac{1}{2 \cdot (n-2)!} - \frac{1}{(n-1)!} + \frac{1}{n!} \right) \epsilon^n \quad (78)$$

Which, for $n = 2$, the coefficient is $\frac{1}{2 \cdot 0!} - \frac{1}{1!} + \frac{1}{2!} = \frac{1}{2} - 1 + \frac{1}{2} = 0$; nonetheless, for $n = 3$, the coefficient is $\frac{1}{2 \cdot 1!} - \frac{1}{2!} + \frac{1}{3!} = \frac{1}{6}$. This shows that the lowest degree term of the power series $P(\epsilon)$ is the cubic.

Finally, plugin the definition $\epsilon := \frac{2b}{a}$, the cubic term becomes:

$$\frac{1}{6} \epsilon^3 = \frac{8}{6} \frac{b^3}{a^3} = \frac{4}{3} \left(\frac{b}{a} \right)^3 \quad (79)$$

(c) Given that $\psi(0) = \frac{1}{\sqrt{\pi a^3}}$, one has the proposed estimate being as follow:

$$P \approx \frac{4}{3} \pi b^3 |\psi(0)|^2 = \frac{4}{3} \pi b^3 \frac{1}{\pi a^3} = \frac{4}{3} \left(\frac{b}{a} \right)^3 \quad (80)$$

Which is exactly as what is proposed in part (b).

(d) Given $b \approx 10^{-15}$ m and $a \approx \frac{1}{2} \cdot 10^{-10}$ m, then one has $\frac{b}{a} \approx 2 \cdot 10^{-5}$. So, one has the following approximation of P :

$$P \approx \frac{4}{3} \left(\frac{b}{a} \right)^3 \approx \frac{32}{3} \cdot 10^{-15} \approx 1.067 \cdot 10^{-16} \quad (81)$$

□